ORACLE

Order Management & Fulfillment

Pricing Advanced Topics



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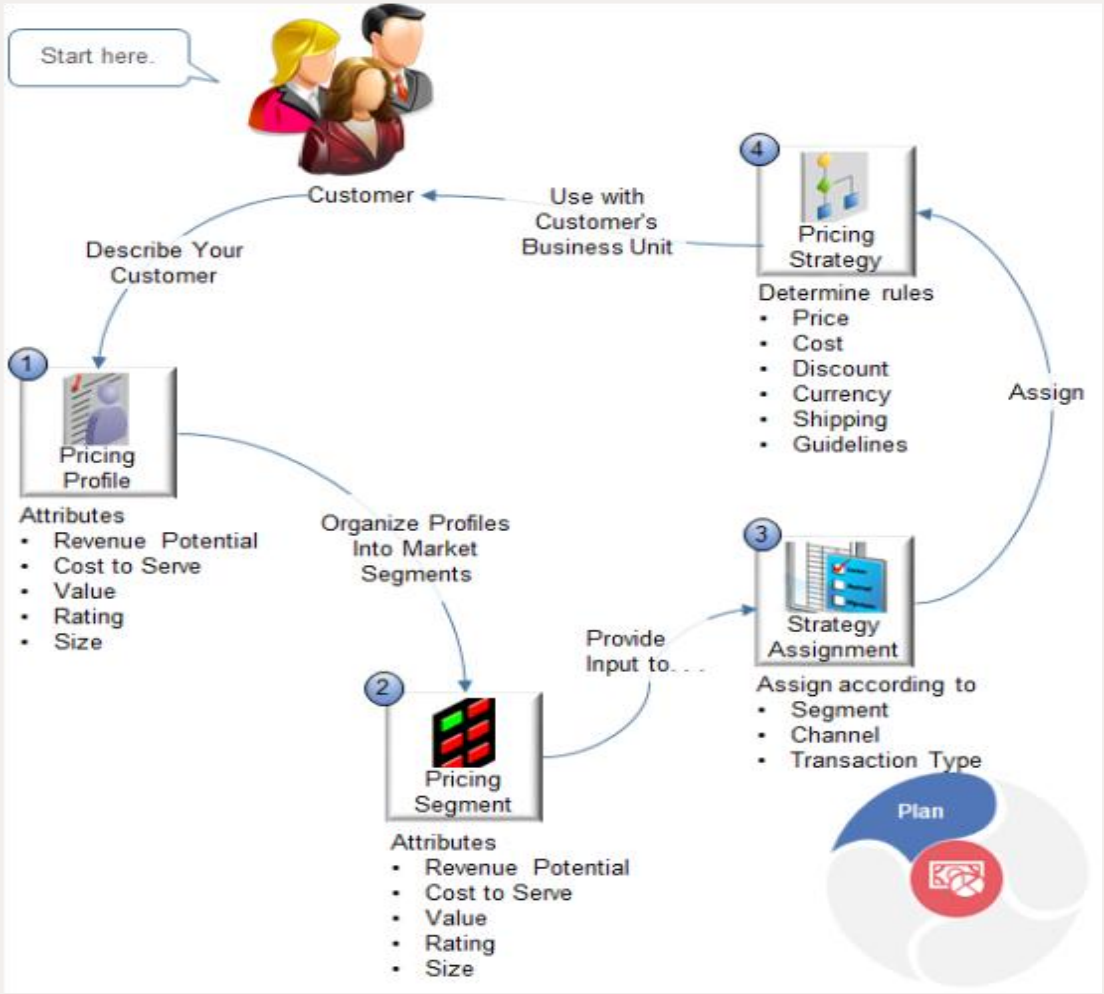
Agenda

- Overview of Pricing
- Pricing Infrastructure Components
- Pricing Matrix
- Service Mappings
- Algorithms
- Use Cases



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How does Pricing work?



Implementation Considerations

Pricing Profile

- Do you need to attach pricing profile to each customer?
- Can we do customer profile updates in bulk?

Pricing Segment

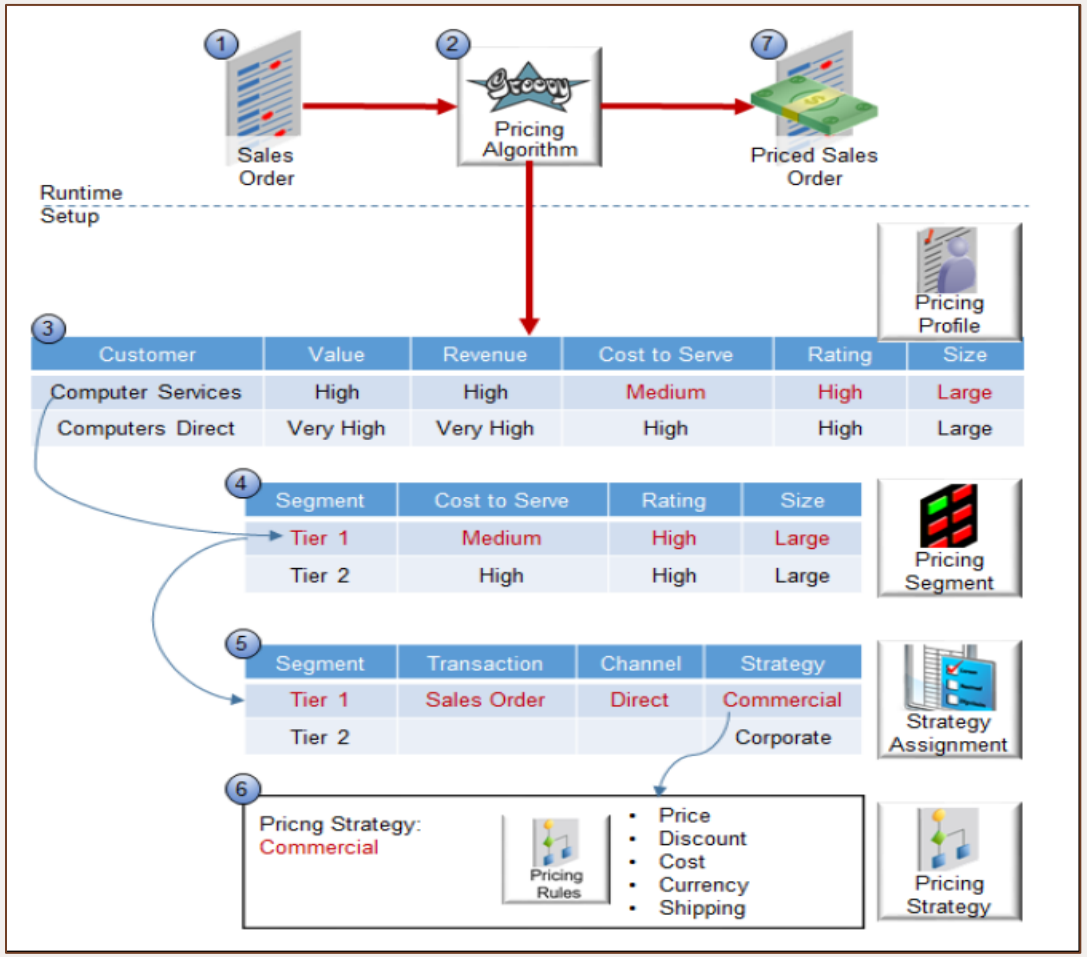
- Is creating a Pricing Segment mandatory?
- Can we extend the List of values for segments like Customer size, cost to serve etc?

Pricing Strategy

- Is creating a Pricing Strategy mandatory?

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How do Pricing Profile, Segment and Strategy work together?



Pricing Profile — Derive the pricing behavior of the customer

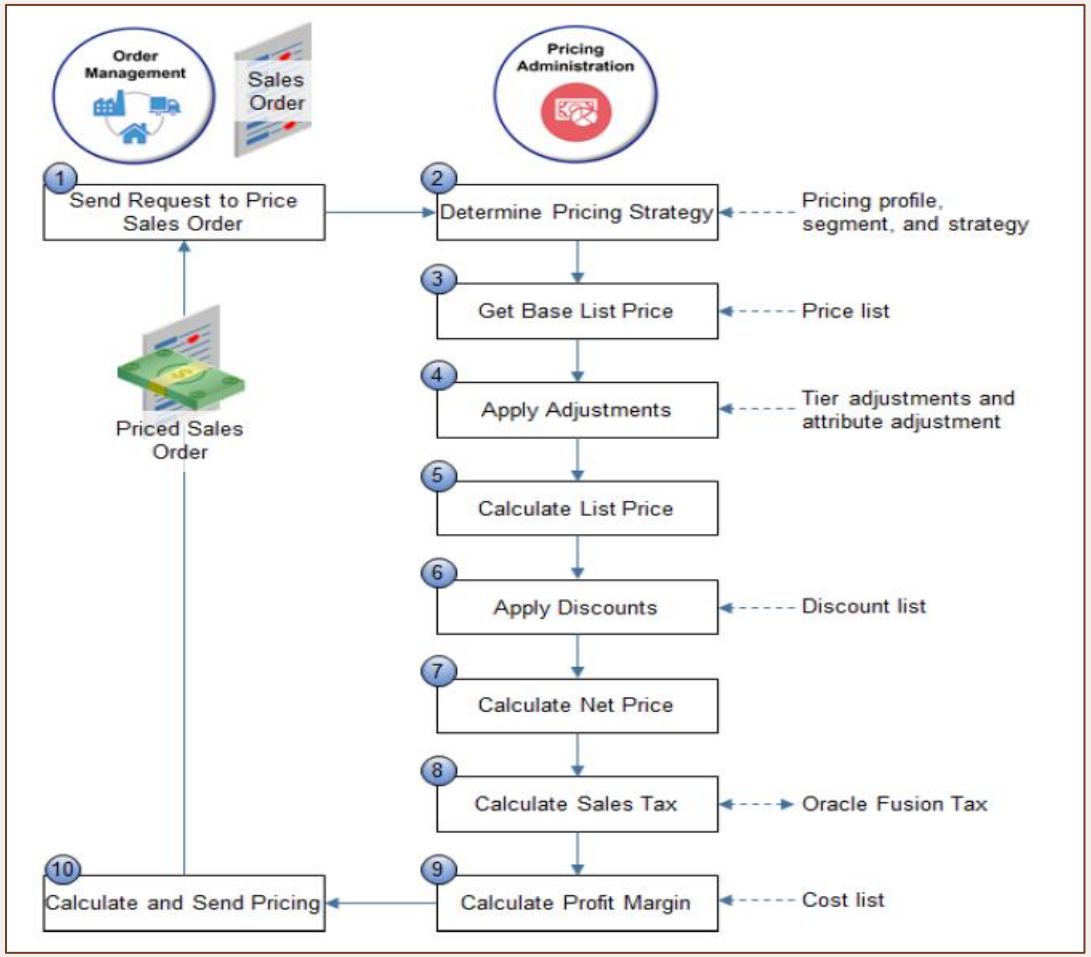
Pricing Segment — Use the pricing segment to group customers who exhibit a similar set of characteristics and buying behaviors together

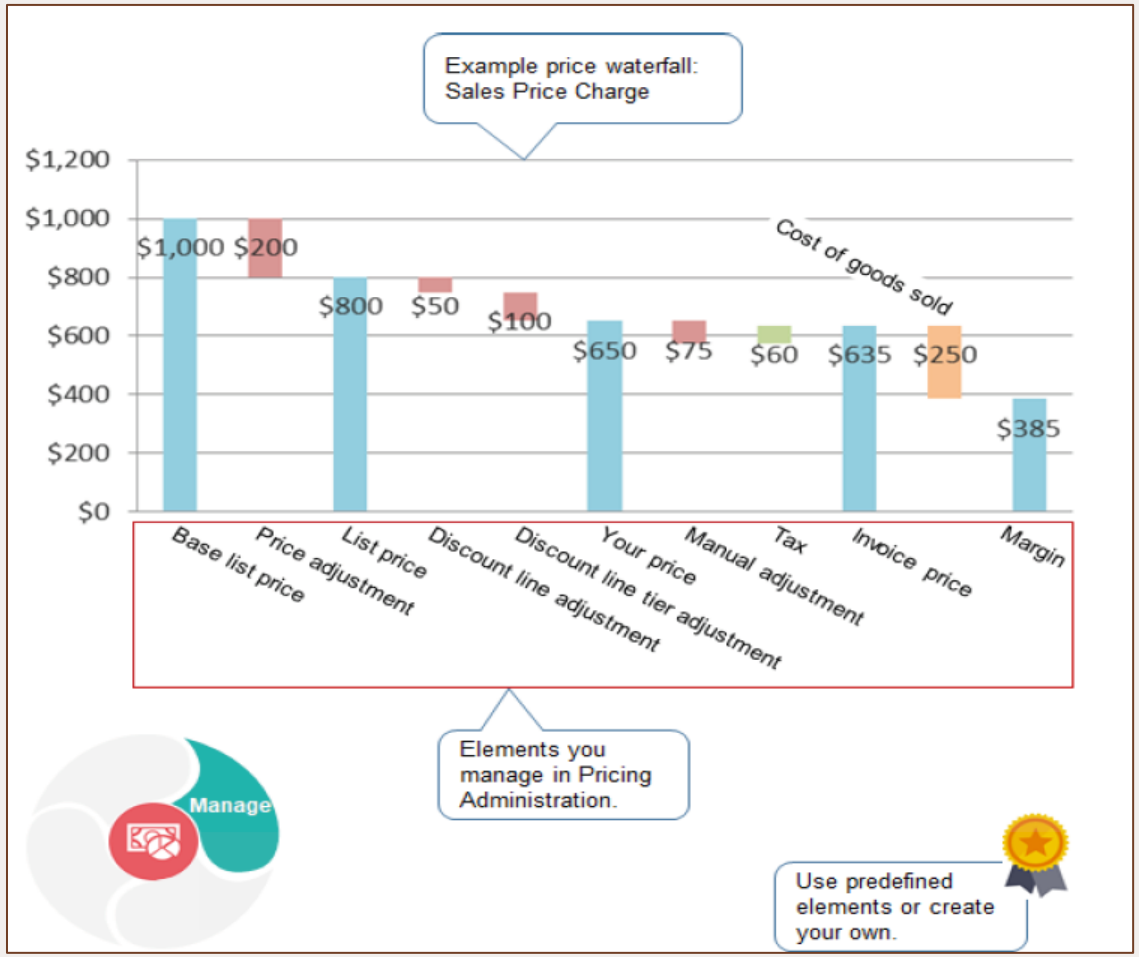
Pricing Strategy — Use different pricing strategies for same pricing segment to support more than one selling scenario.

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How is the final price derived?

Price components





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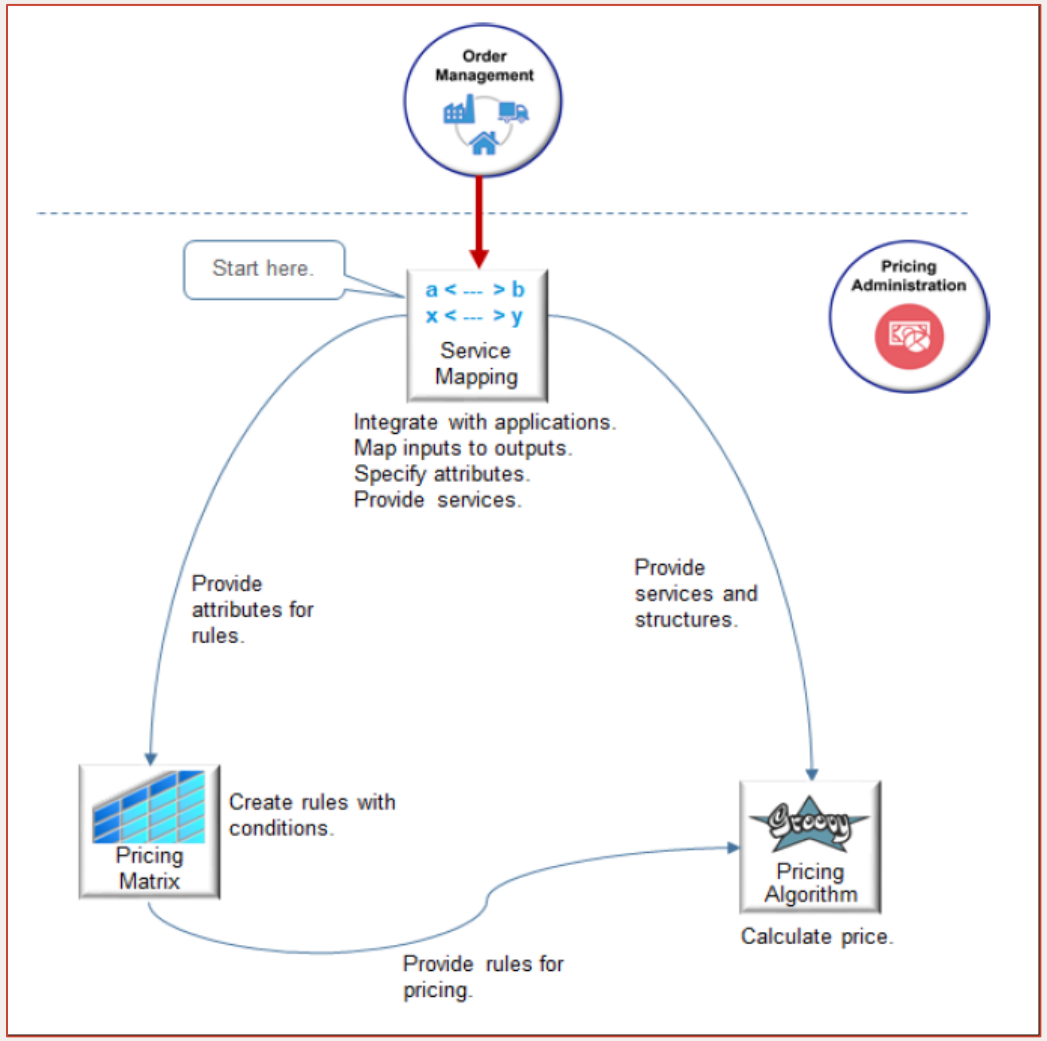
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Pricing Execution Components

Service Mappings	• Canonical Representation of data
Matrices	• Multi-dimensional rules with condition and results in a table format
Algorithms	• Data manipulation via dynamic document element relationship

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Pricing Infrastructure Components



Service Mappings:
Integrate with the calling application that requires pricing, such as Order Management Cloud. Map inputs from the calling application to outputs that the matrix and algorithm can use.

Set up attributes so rules in the pricing matrix can use them. A service mapping provides the compare-to attributes in the matrix class.

Set up services and pricing entities to provide structure for variables that the pricing algorithm uses.

Pricing Matrix:
Set up conditions and dimensions in the pricing matrix.

Choose attributes in the condition and result columns of the matrix class from the entity attributes you set up in the service mapping.

The algorithm uses condition dimensions and their values to compare the runtime value against the value of the design time rule. The result values are the result of evaluating the conditions.

Pricing Algorithm:
Set up the algorithm so it uses the pricing matrix to evaluate and apply pricing rules at run time.

Set up the algorithm to transform the attribute values it receives from

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Pricing Matrix

Use a pricing matrix to apply conditional logic on a pricing entity according to the value of an attribute that you specify.

- Matrix provides ability to define multi-dimensional rules
- Each Matrix can have condition and result dimensions.
- Each matrix has a set of rules. Rules can have start and end dates.
- The source for the values of the dimension is derived from the 'Domain' definition
- Matrixes are 'Dynamic' because the Matrix structure is not fixed

Sales Pricing Strategy Assignment Matrix Rules

	Condition Columns		Result Columns
	Pricing Segment (=)	OrderType (=)	* Pricing Strategy
Rule1	DefaultSegment	Standard Orders	Default Pricing Strategy
Rule2	Corporate Segment	Standard Order Models	QP Competitive Price Strategy
Rule3	Corporate Segment	Standard Orders	Corporate Pricing Strategy

Conditions

Domain

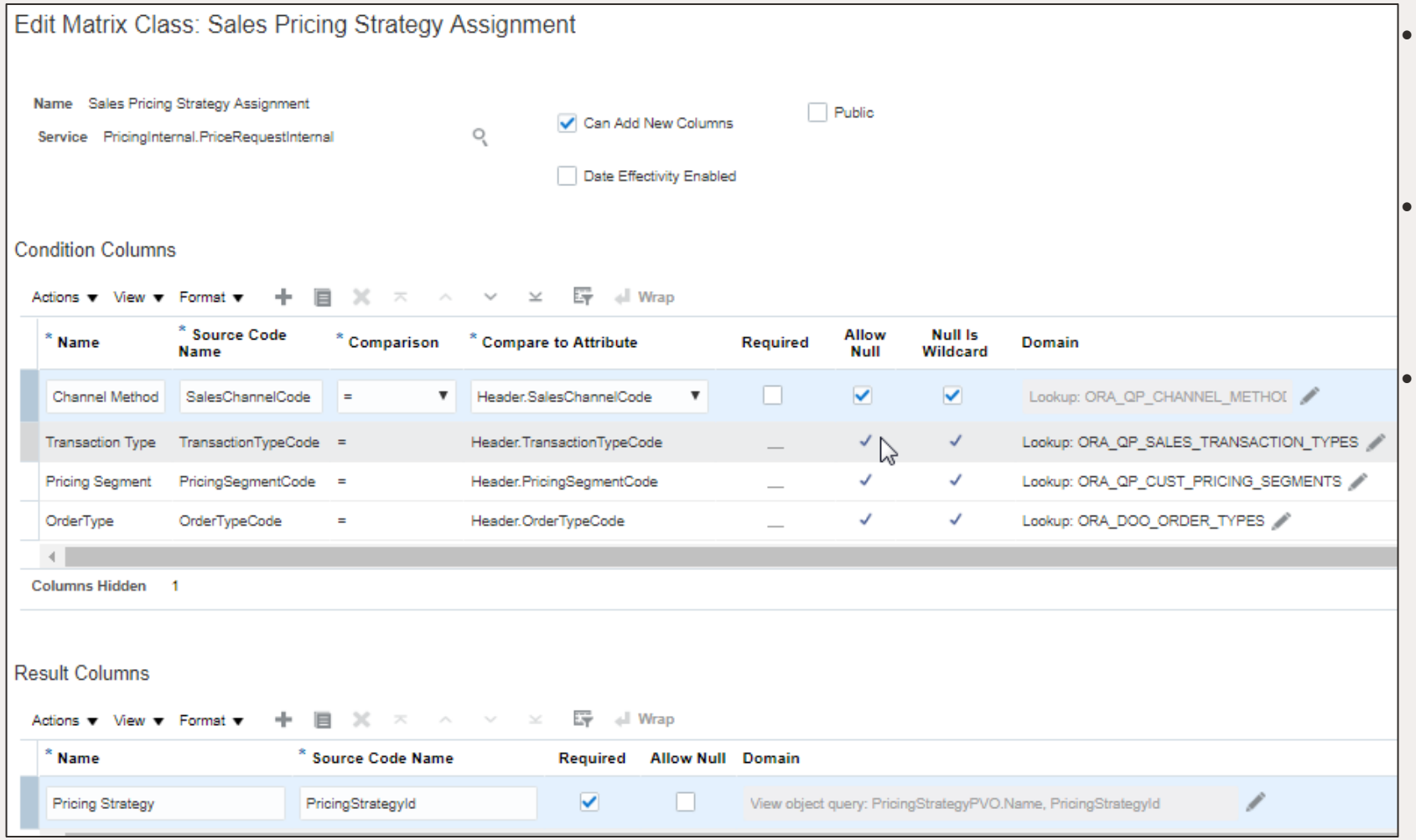
- Standard Orders
- Dropship Orders
- Mixed Orders
- Return Orders
- Standard B2B Products
- Standard Order Models
- Standard Orders - Finished Goods

Results

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Pricing Matrix Class

Matrix Class captures the structure of a Matrix including the Condition and Result Dimensions



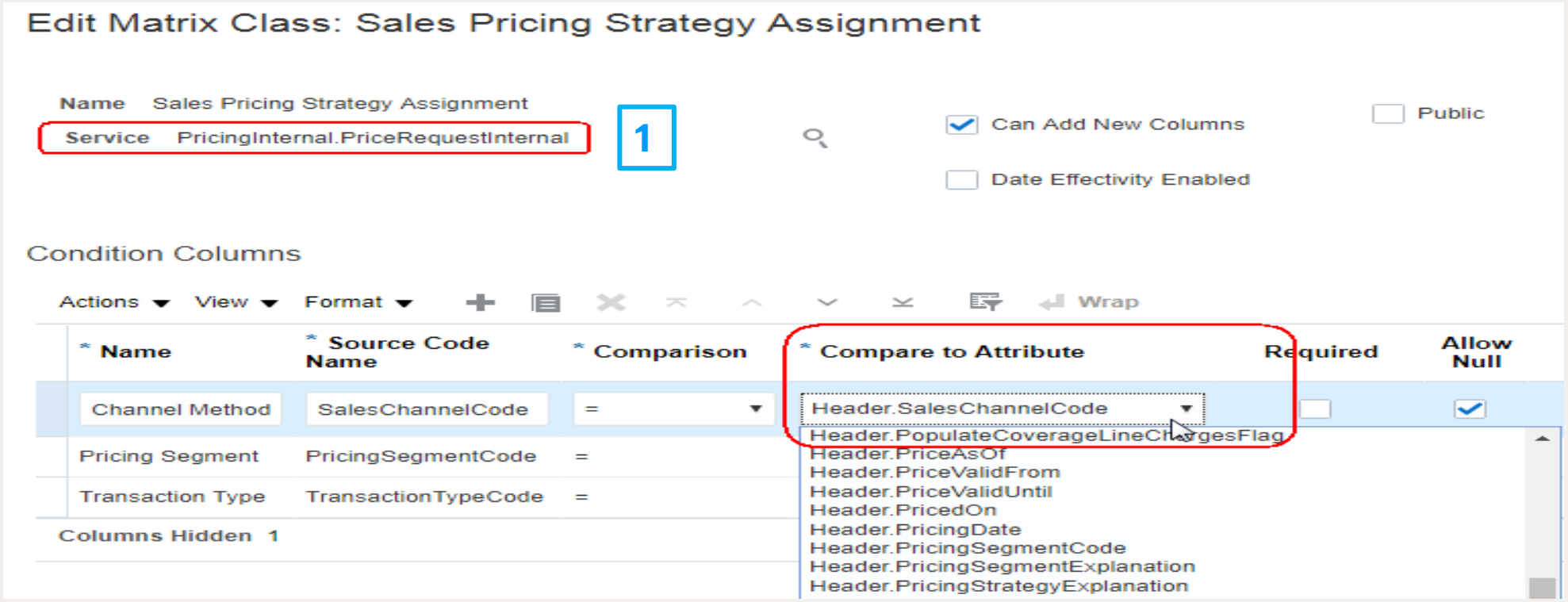
- Matrix Class Key Attributes
  - Service
  - Date Enabled
  - Can Add New Columns
- Dimension Admin Controls
  - Required
  - Allow Null
  - Domain
- Dimension Runtime Controls
  - Compare To
  - Comparison
  - Null Is Wildcard

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Matrix Classes and Dimensions

Service and Compare to Attribute

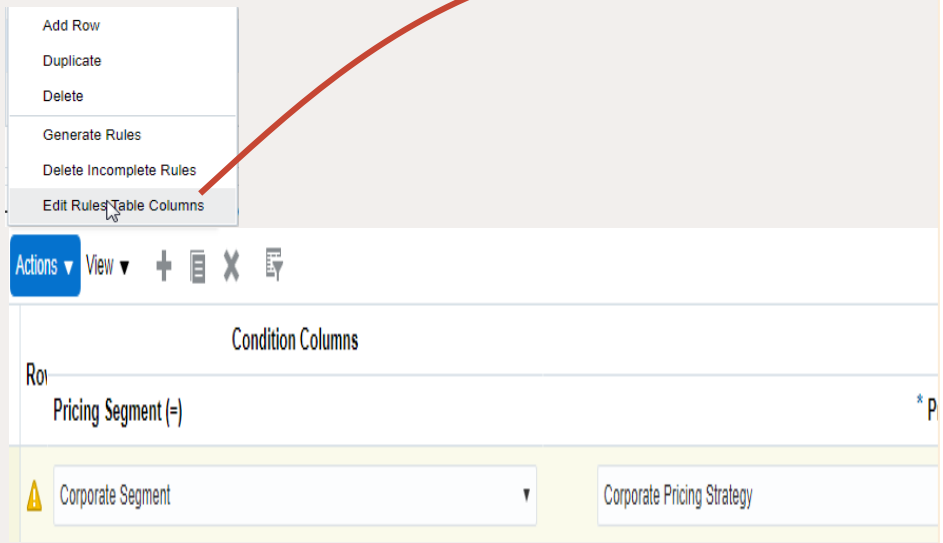
1. Service in Seeded Matrix Classes is PricingInternal.PriceRequestInternal
2. Service definition in Service Mappings provides the list of Compare to Attributes

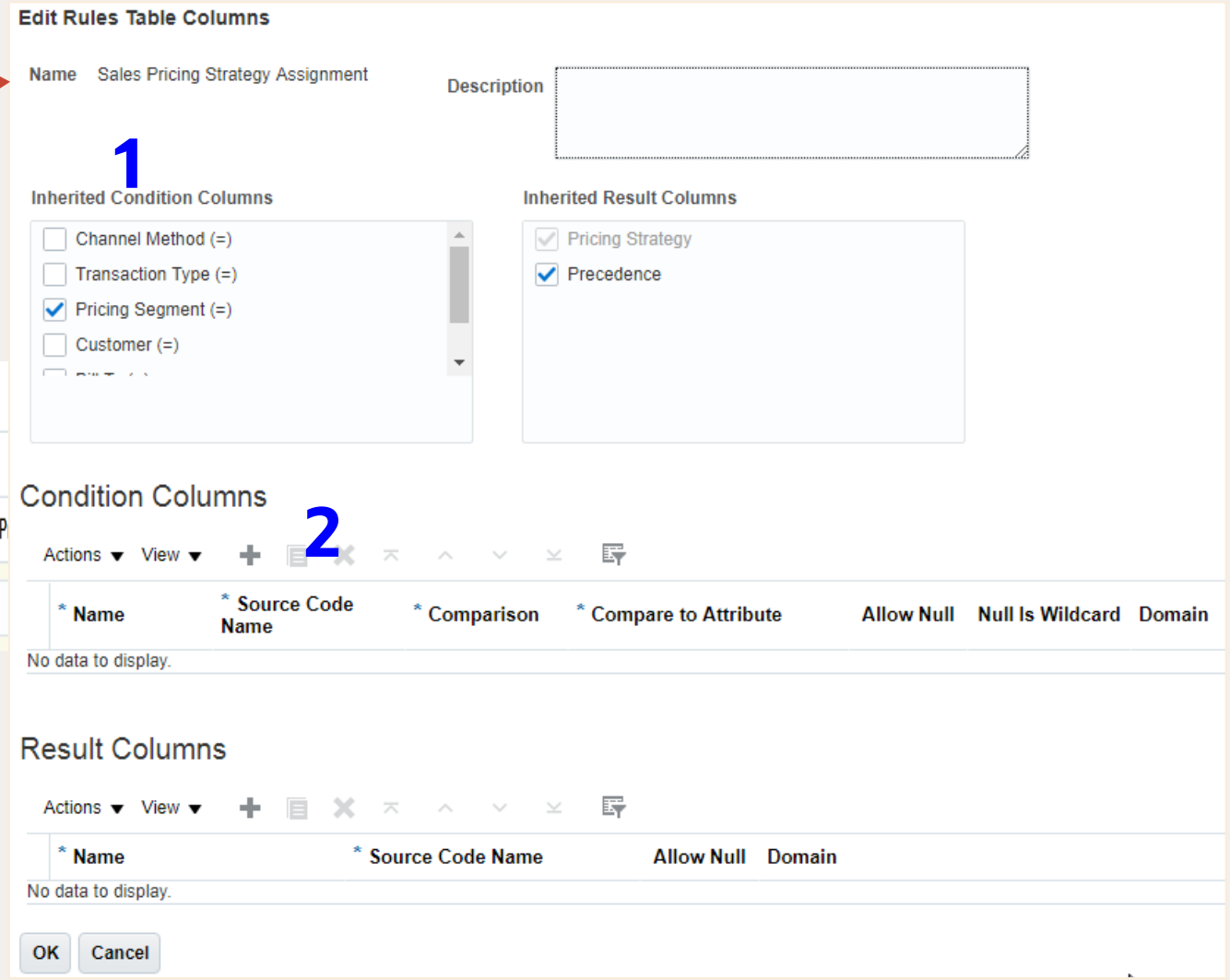


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Adding Dimensions to a Matrix

2 ways to add dimensions to matrixes:



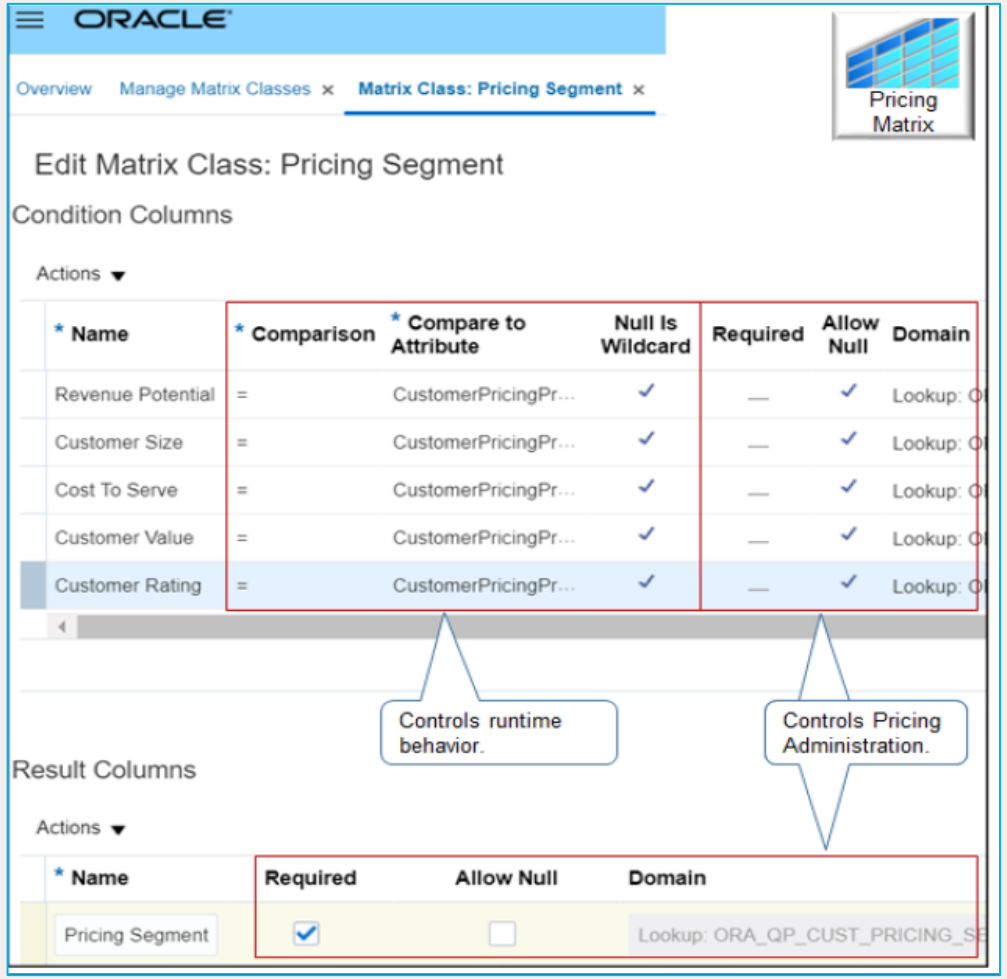


1. Dimensions inherited from the Matrix Class

2. Custom Condition and Result Dimensions added to the Matrix

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Pricing Matrix Columns



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Implementation considerations

Matrix classes are mapped to a matrix type. The mapping ensures that the correct matrix class is leveraged by the applicable entities in the Pricing Administration UI, as well as during price execution.

Oracle Fusion Pricing comes predefined with matrix types and matrix classes. Each type references one, and only one, class. Each class specifies conditions and results.

Matrix Type	Algorithm	Step
Pricing Segment	Get Sales Pricing Strategy	Derive Pricing Segment
Sales Pricing Strategy Assignment	Get Sales Pricing Strategy	Get Header Strategy
Price List Charge Adjustment	Apply Matrices	Evaluate Pricing Matrices
Discount Adjustment	Apply Matrices	Evaluate Pricing Matrices
Pricing Term Adjustment	Apply Matrices	Evaluate Pricing Matrices
Line Strategy	Get Line Pricing Strategy	Get Pricing Strategy
Pricing Charge Guideline	Apply Matrix Rules for Guidelines	Evaluate Pricing Matrices

- ✓ Use the predefined classes, when possible
- ✓ If you must modify a predefined class, or add a new one:
  - You typically define a new condition column.
  - You only rarely need to define a result column.
  - You can use only a predefined result column with a predefined pricing algorithm.
  - Don't remove a predefined result column. It might cause an error when the pricing algorithm runs.
  - If you add a new result column, then you must modify the pricing algorithm in the same way to make sure the pricing algorithm uses the new result.

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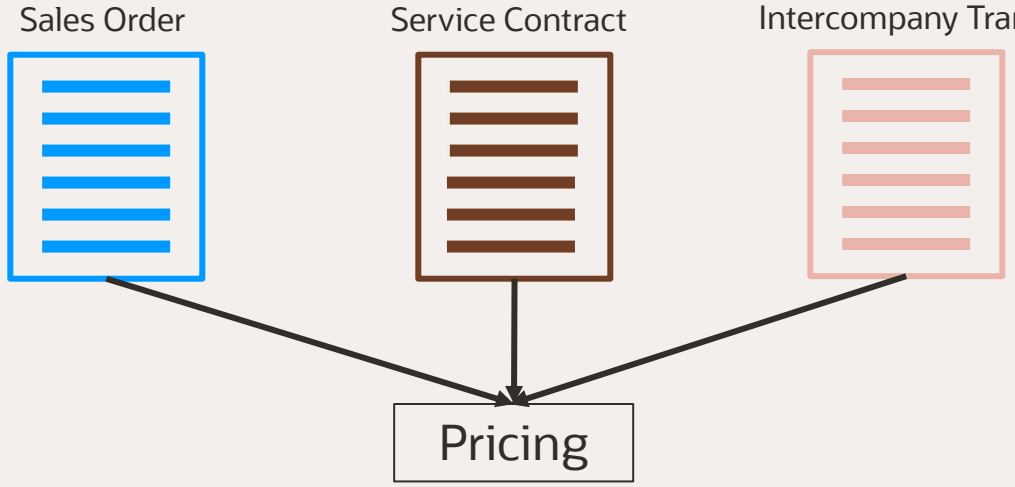
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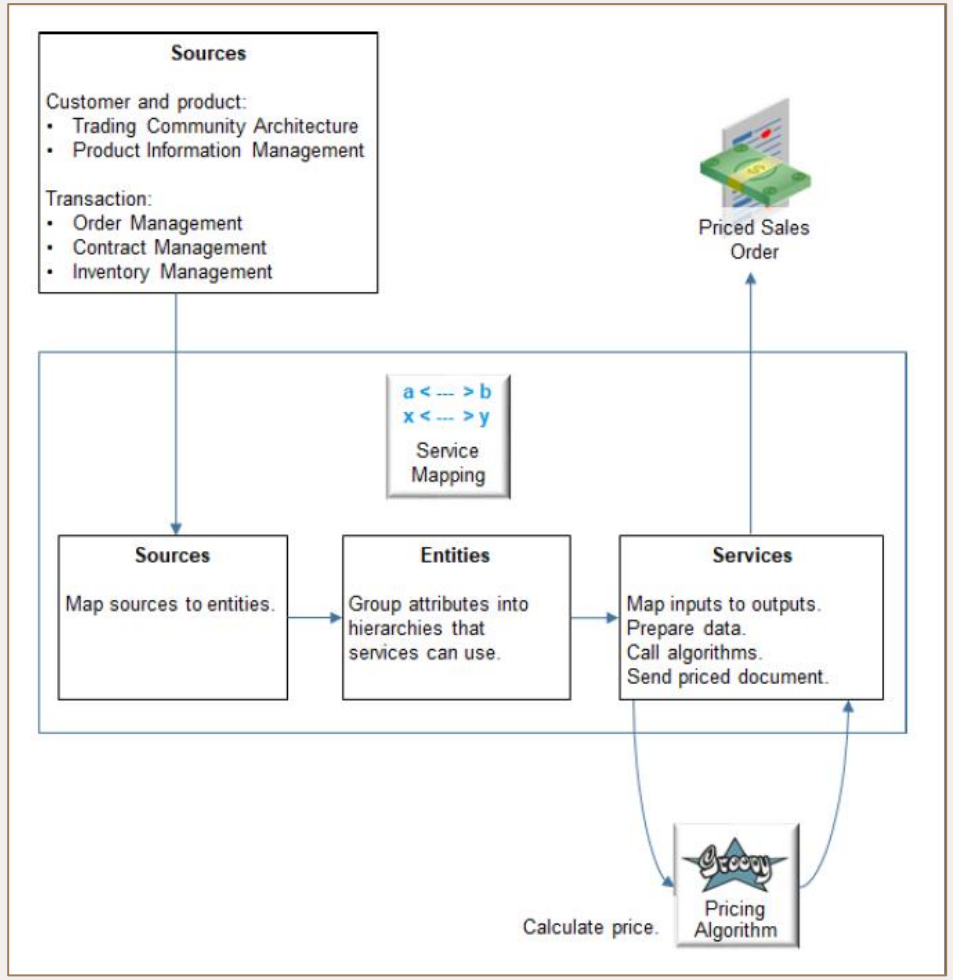


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Service Mapping

A service mapping is a map that creates relationships between services, sources, entities, and attributes. It specifies a group of pricing entities and attributes that a service can retrieve and update.

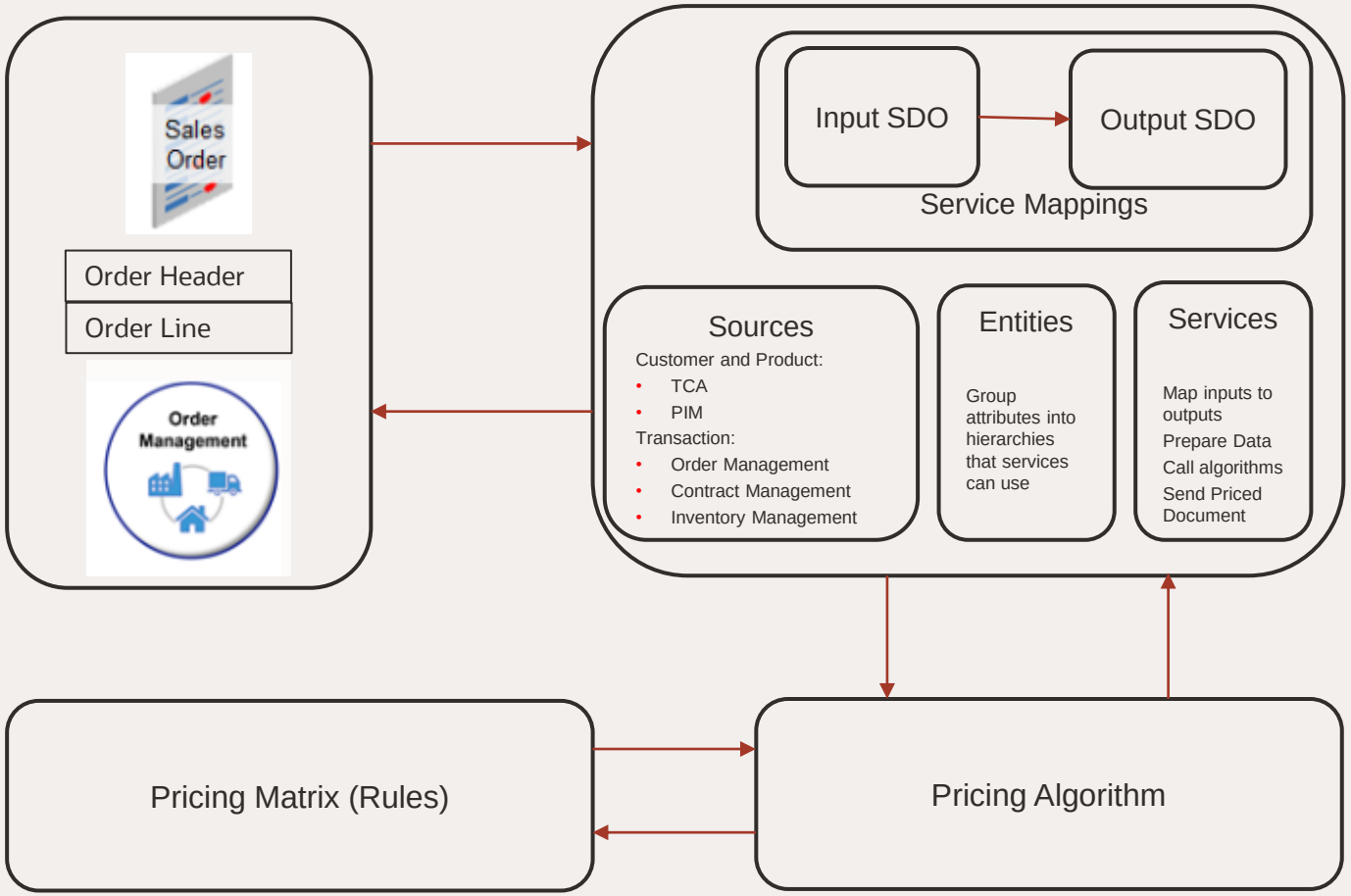




- Similar Pricing Requirements
- Consistent Pricing
- Similar attributes to drive pricing
- Attribute-based pricing rules
- Extensibility Requirements

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Pricing Entities



SDO (Service Data Object):
A data structure that doesn't conform to a single programming language. It facilitates communication between applications. It allows a service to call another service without having to conform to the requirements of a single language. A service data object typically includes elements arranged in a tree structure that includes a root and branches, and it allows the calling service to access the data that these elements reference.

Source:
The source is a mapping that specifies the relationship between the source and entities. It provides a structure so Pricing can model data in the input SDO.

For example, the predefined Sales service mapping includes the Order Total source, and this source includes details about the Order Header and Order Line.

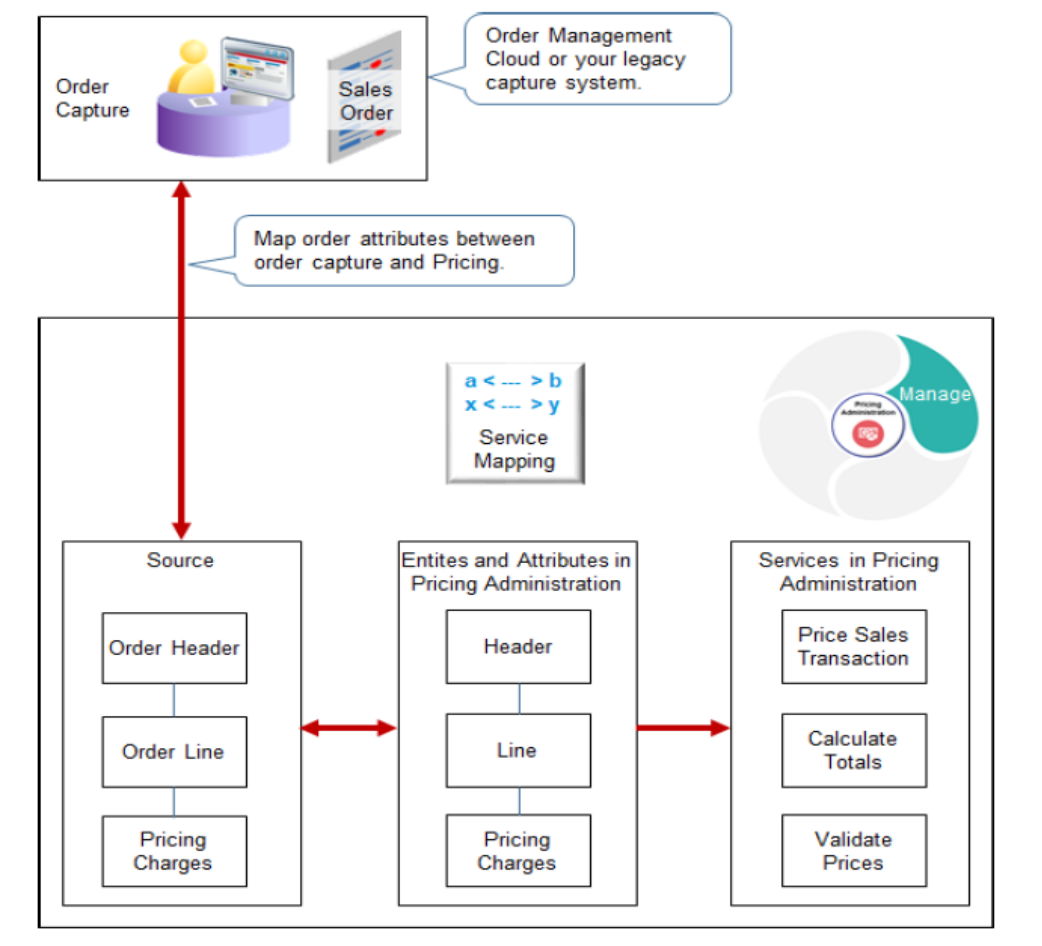
Maps between the OrderLine source and the Header entity
Maps between the OrderLine source and the Line entity

Service:
Pricing uses these Services to price a sales order. Services can be internal or external.

Entities:
Reference the entities that the Service Mapping requires to structure the output SDO

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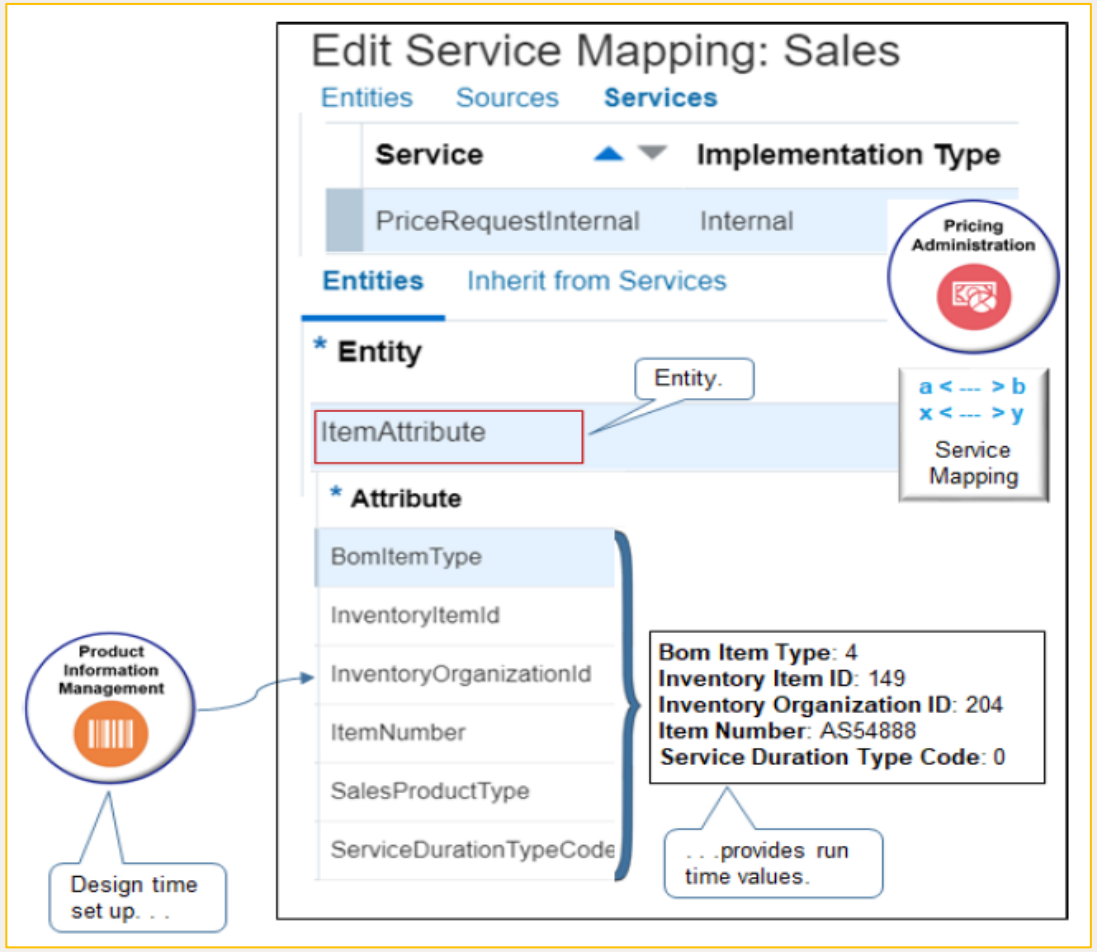
Service Mapping for Orders



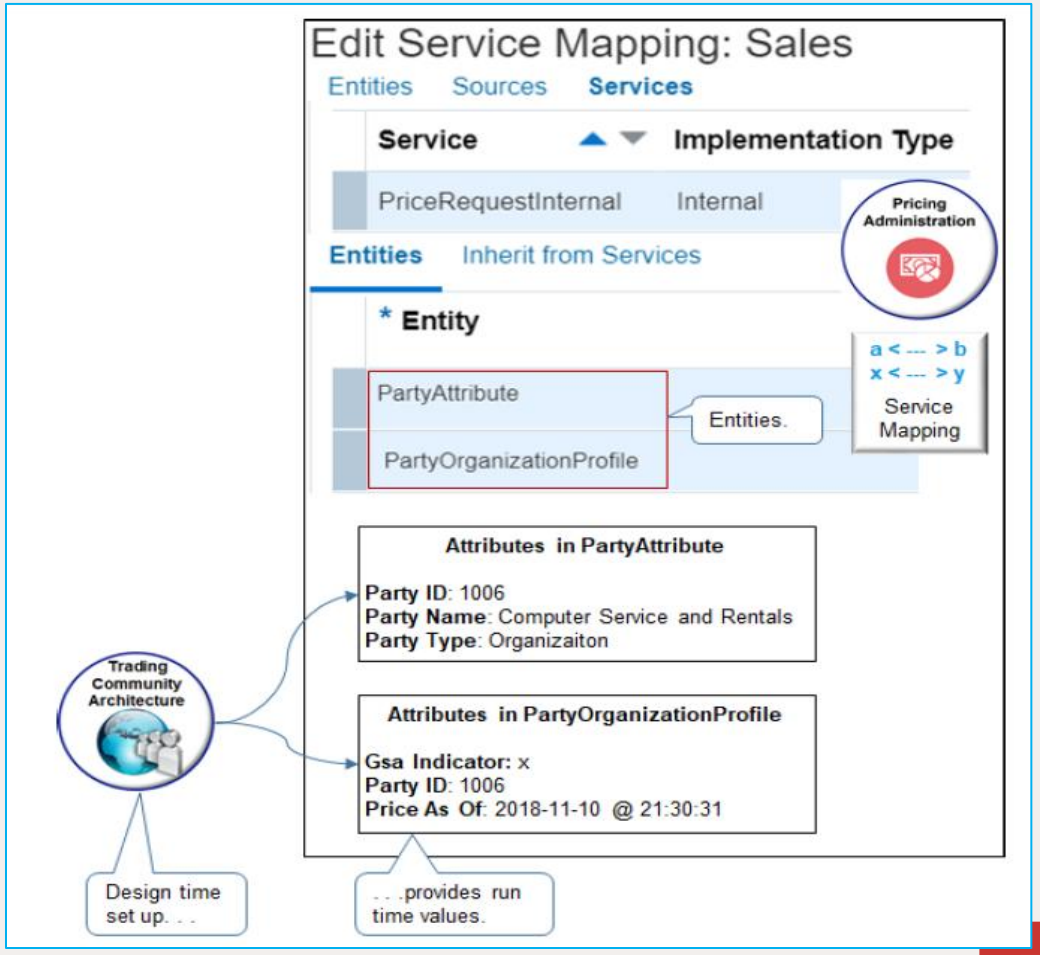
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Service Mapping Entity - Examples

Pricing uses the ItemAttribute entity to store data about the item you need to price.



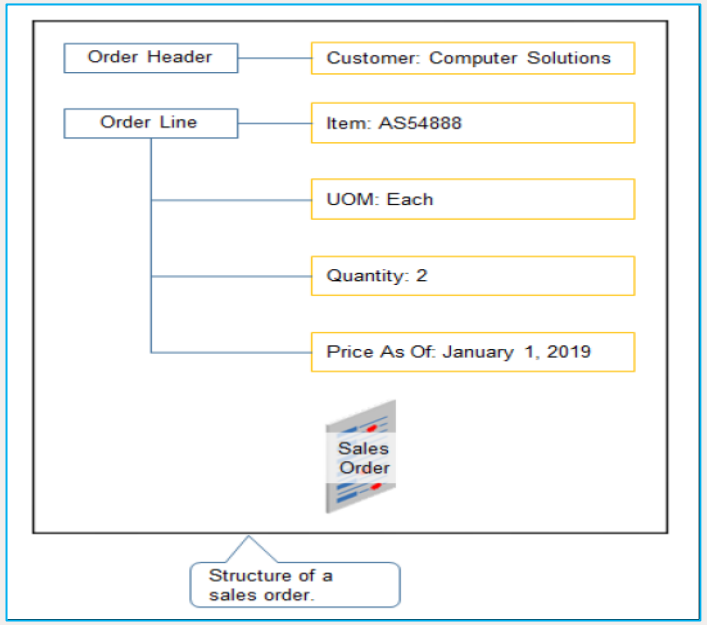
Pricing gets this customer data, from Trading Community Architecture.

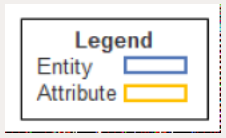


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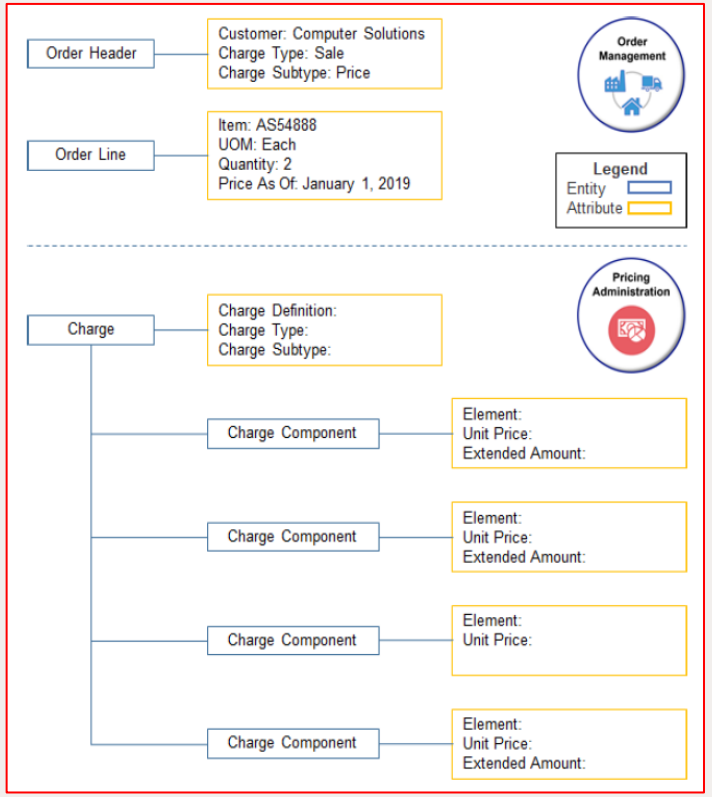
Service Mapping - SDOs

Input SDO

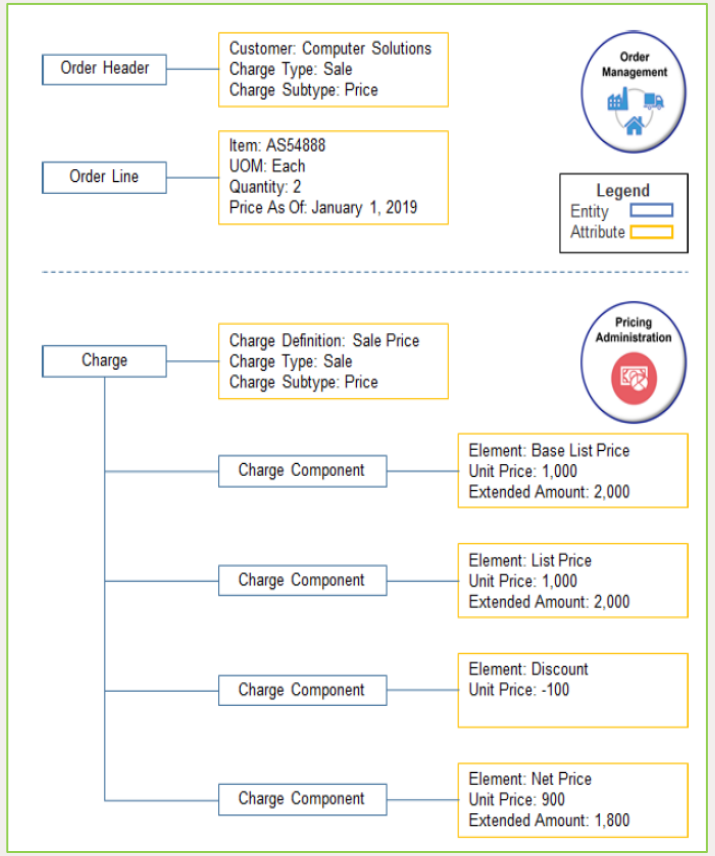




Output SDO



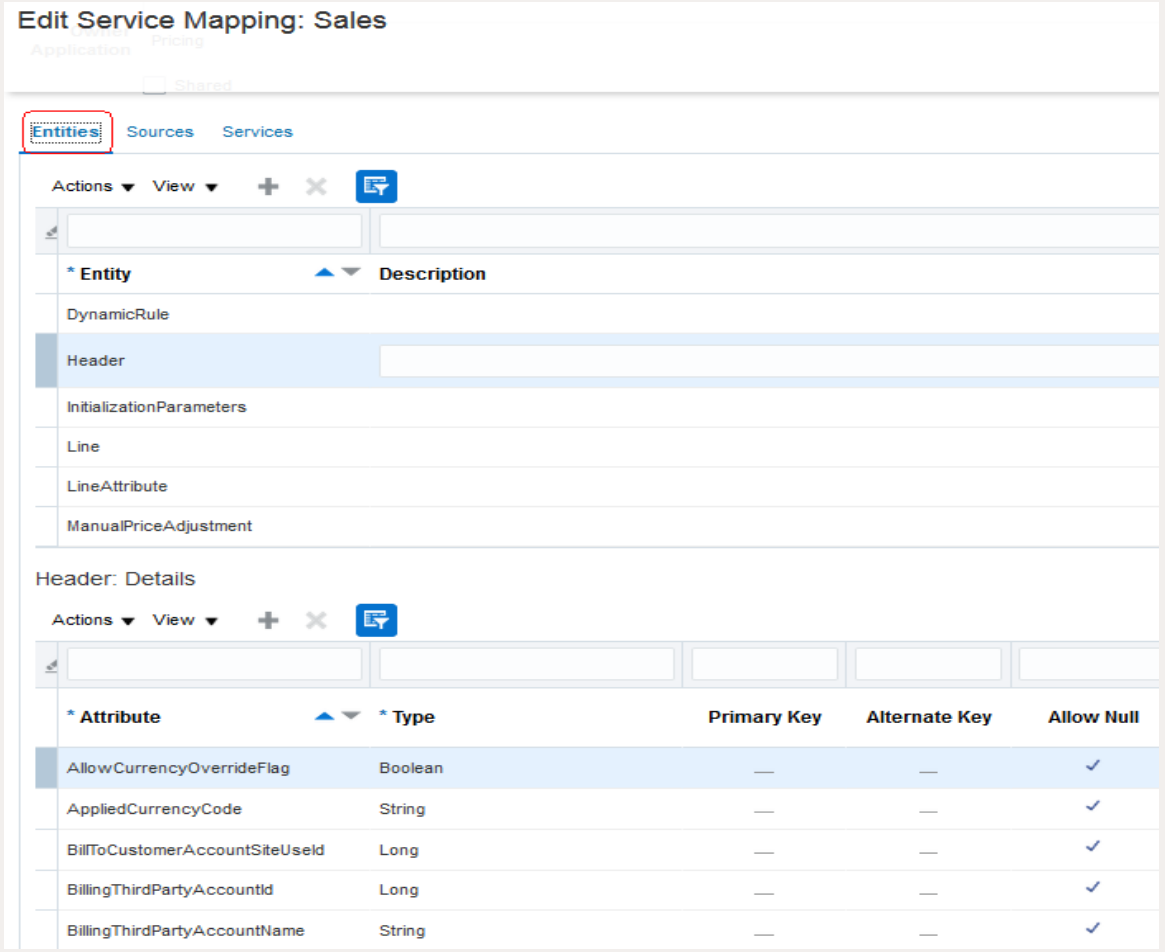
Output SDO with Pricing data



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Service Mapping – Entities and Attributes

- Canonical Representation
  - Key Entities
    - Header
    - Line
    - Charge
    - Charge Component
- Customers can create new entities and attributes



Key Properties

- Entity/Attribute – Unique Name, _Custom suffix
- Description
- Type - Indicates the data type of the attribute
- Read - Indicates whether the attribute will be populated on the input service SDO.
- Write - Indicates whether the attribute and be populated on the output service SDO.
- Description - A description of what this attribute represents

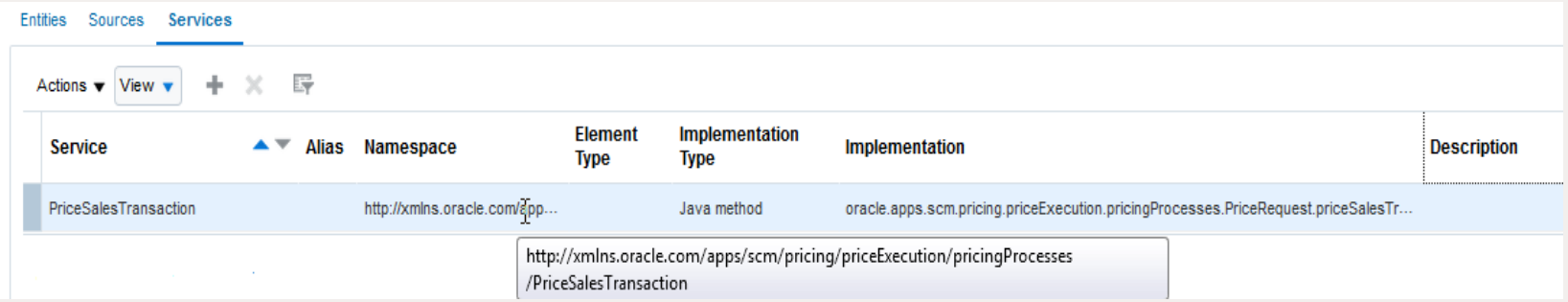
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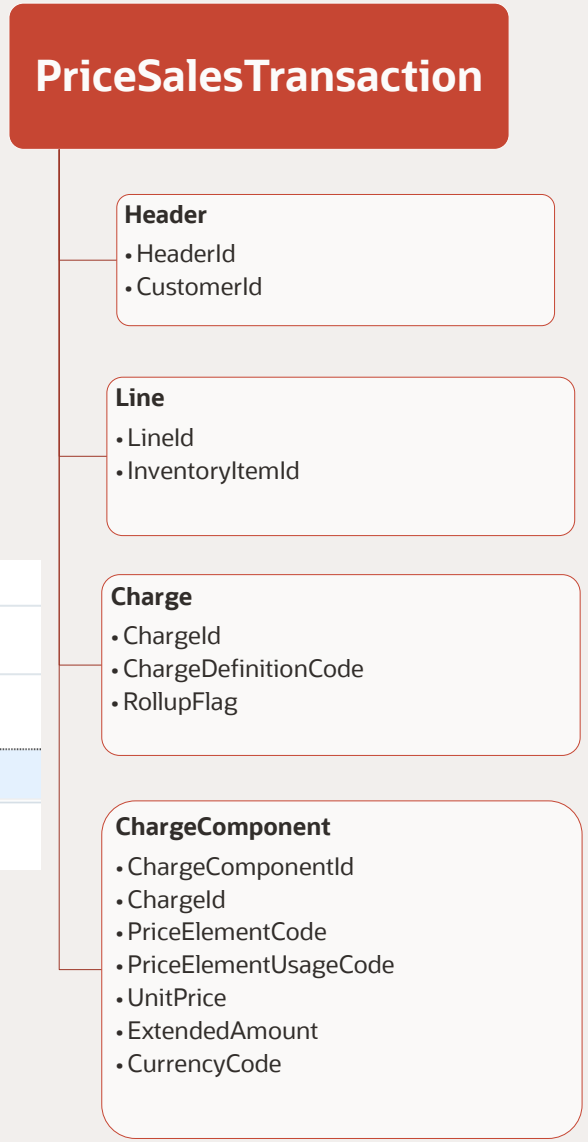
Service Mapping – Services

Services are operations performed on the Sales Object

Services define:

1. The implementation to invoke – e.g. java method or an algorithm.
2. The schema for the input and output - subset of the canonical representation





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Implementation considerations

Service Mapping Edits require Sandbox creation to add the required attributes and make edits. Following are some of the use cases from an implementation standpoint.

- ✓ Assign Pricing Strategy According to Order Type
- ✓ Set Up Discounts That Accumulate or Cascade
- ✓ Set Up Price Lists for Each Customer
- ✓ Use Extensible Flexfields with Pricing
- ✓ Adjust Price According to Predefined Attributes
- ✓ Override Net Price

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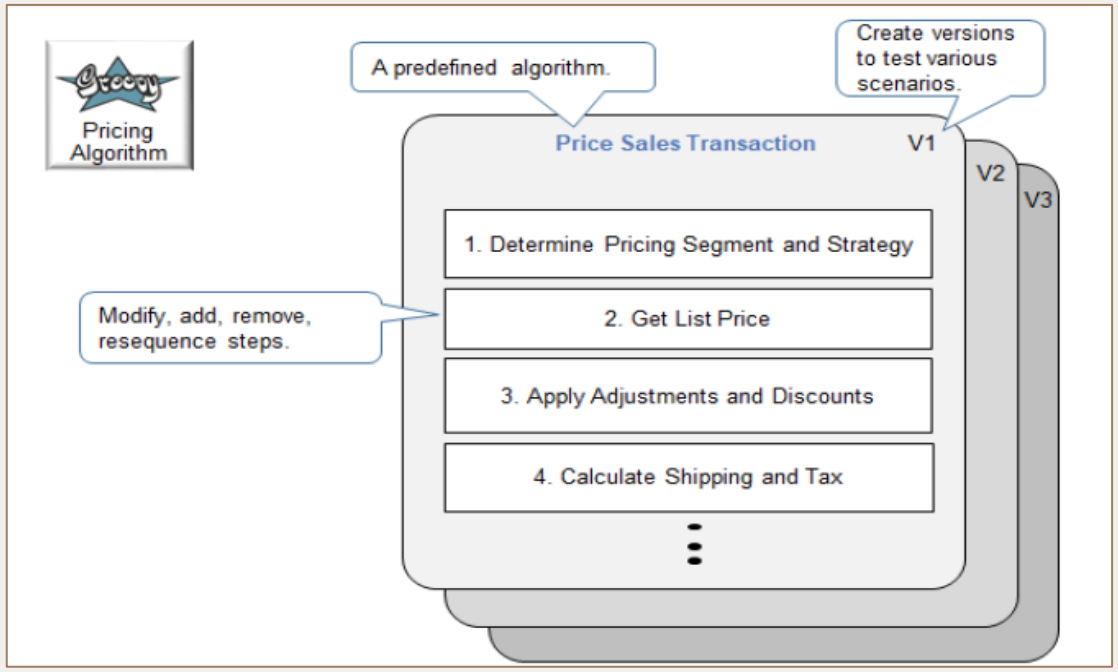
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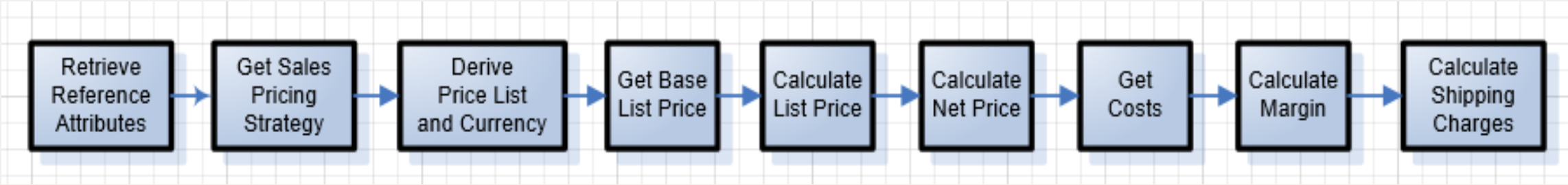


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Algorithm Overview

- A pricing algorithm is a set of rules that uses conditional logic, variables, functions, and Groovy script to manipulate data that affects pricing. Use it to modify the logic that Oracle Fusion Pricing uses when it prices an item.
- Extensibility to address dynamic pricing requirements
- Pricing implementers control publishing extension versions
- No Dependency on Cloud Operations and System Administrators





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Algorithm Attributes

- Unique Name - Upper Camel Case with Spaces, Custom Suffix
- Version – Zero and 1 are seeded. Higher versions are customer extended versions
- Status – In Progress, Published, Inactive
- Public – No, Use Yes only if same algorithm is to be used across applications eg. Contracts or OM
- Start/End Dates – Effectivity based on the Algorithm Execution Date.
- Description
- Owner Application – Indicates the owner of the algorithm eg. Pricing
- Module – Module for the algorithm eg. Price Execution

Name	Version	Status	Public	Start Date	End Date	Description	Owner Application
Retrieve Reference Attributes	2	In progress	—	1/26/19 7:11 PM		This step retrieves the item and customer reference attributes.	Pricing
Retrieve Reference Attributes	1	Published	—	12/26/18 1:19 PM		This step retrieves the item and customer reference attributes.	Pricing
Retrieve Reference Attributes	3	Inactive	—	1/26/19 8:15 PM		This step retrieves the item and customer reference attributes.	Pricing
Retrieve Reference Attributes	0	Published	—	11/7/17 11:13 PM		This step retrieves the item and customer reference attributes.	Pricing

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Algorithm Components - Variables

Variables have:

Name – Unique name within that algorithm

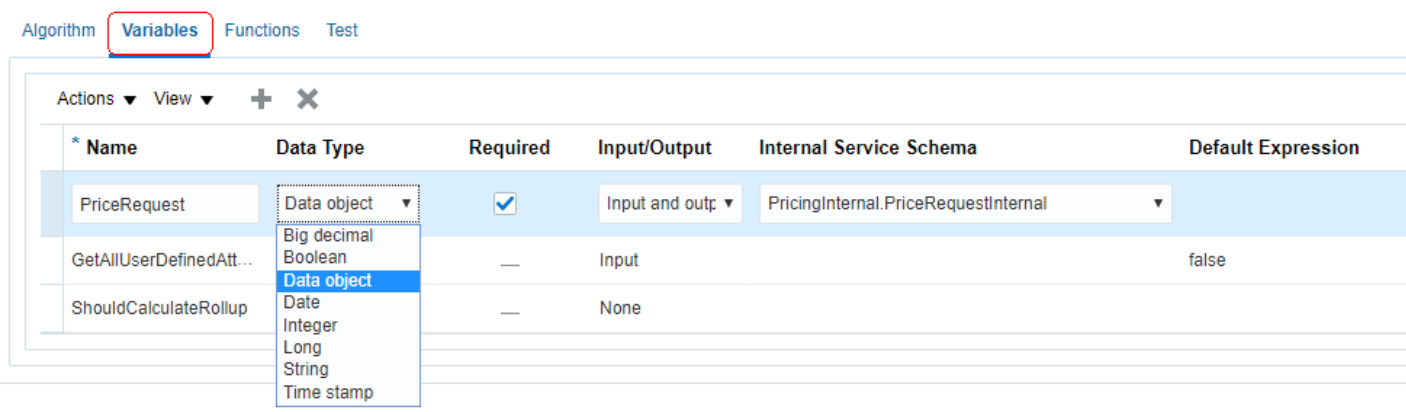
Data type – One of the types listed above. The Data Object type is based on Service Mappings Services.

Required – Indicates that a value is required for executing the algorithm

Input/Output – Indicates whether the variable is input only or output only or both input and output.

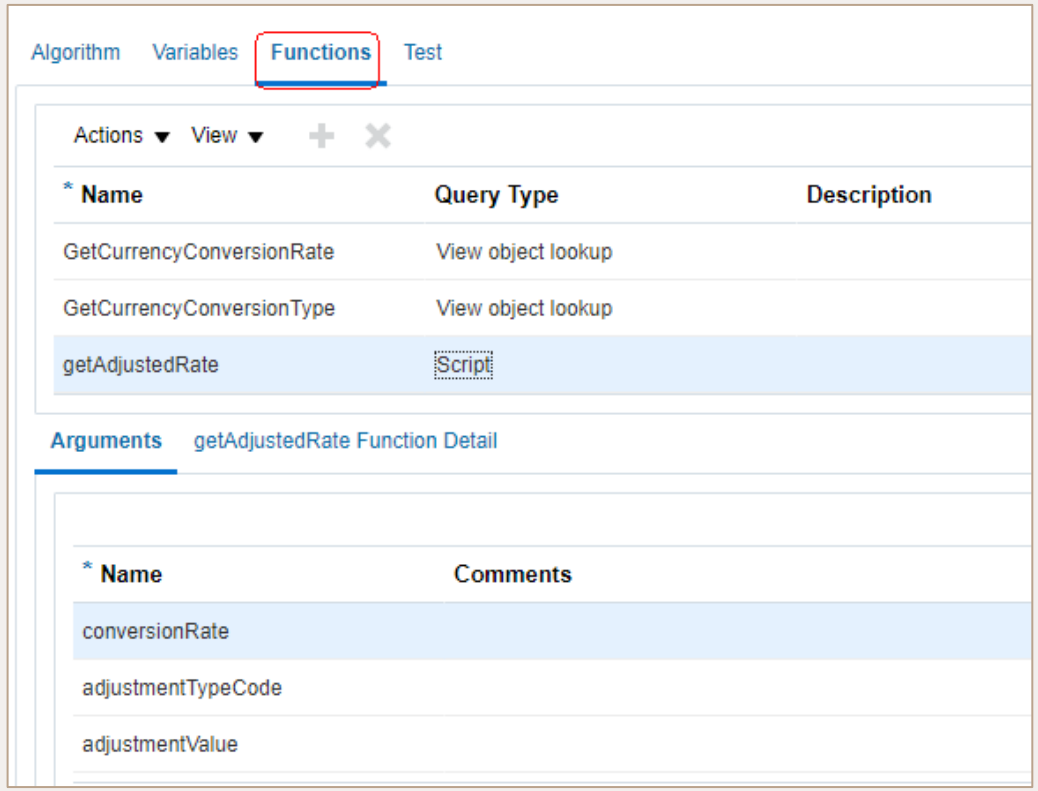
Internal Service Schema – Applicable to Data Objects. The schema is one of the services from the Service Mappings. Most algorithms use at least one Data Object Variable.

Default Expression – Default values can be provided using groovy expressions for all data types except



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Algorithm Components - Functions



Function Types

1. Script – Perform procedural logic and return results
2. View Object Query – Lookup data and return results

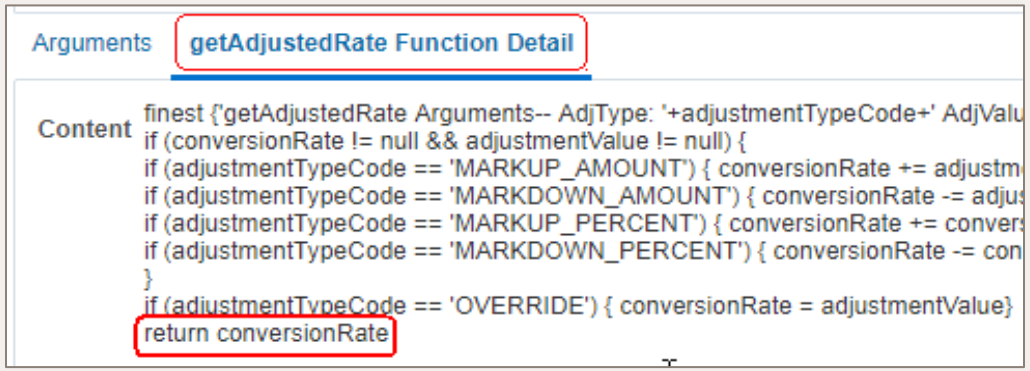
Function Arguments

- Unique Name
- Comments

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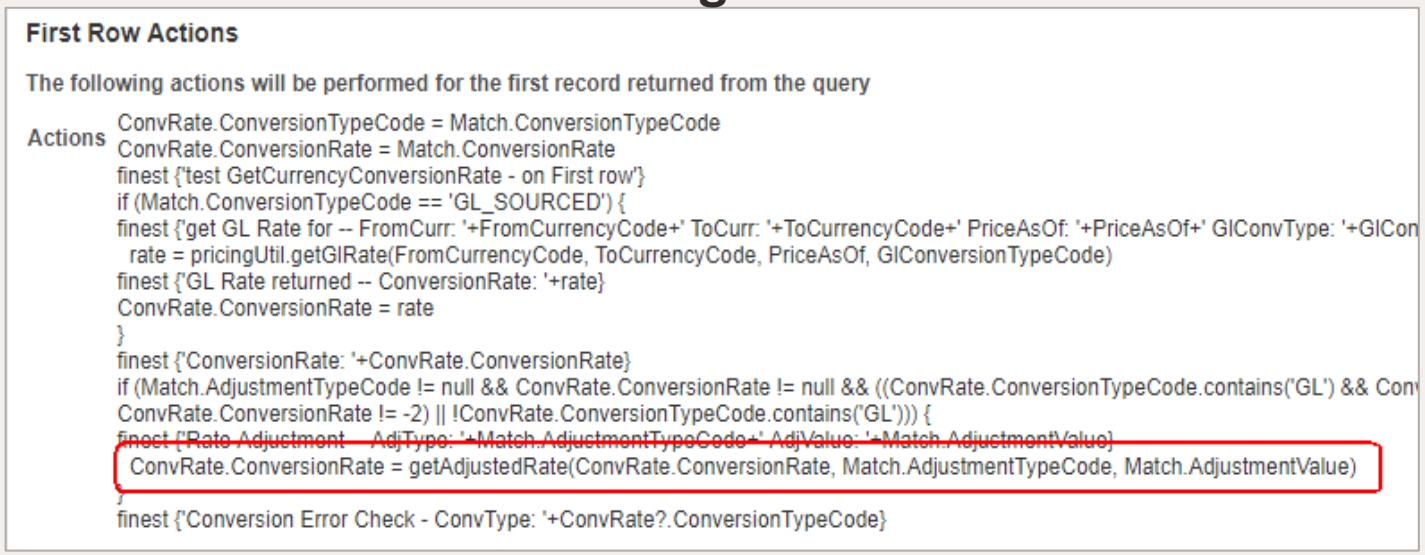
Algorithm Components - Functions

Function Detail



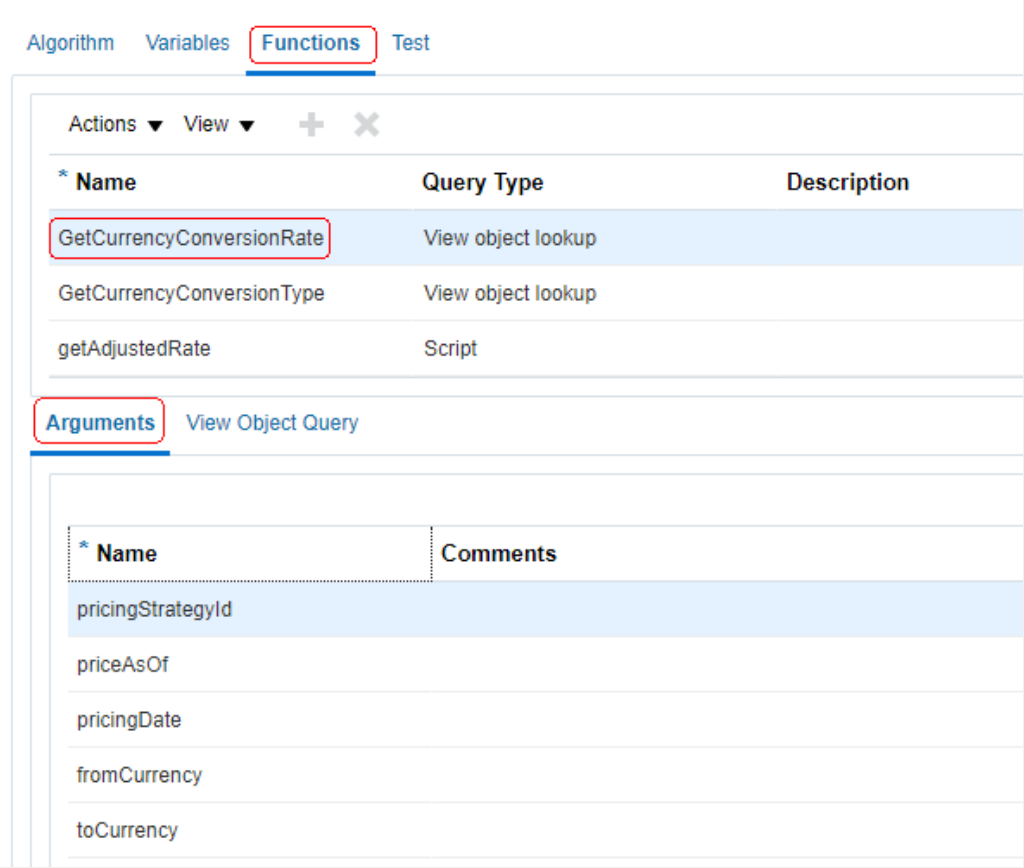
- Function takes 3 input arguments
- Returns output value

Function Invocation in the algorithm



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Algorithm Components – Functions View Object



Function Name

GetCurrencyConversionRate

Arguments

pricingStrategyId

priceAsOf

pricingDate

fromCurrency

toCurrency

businessUnit

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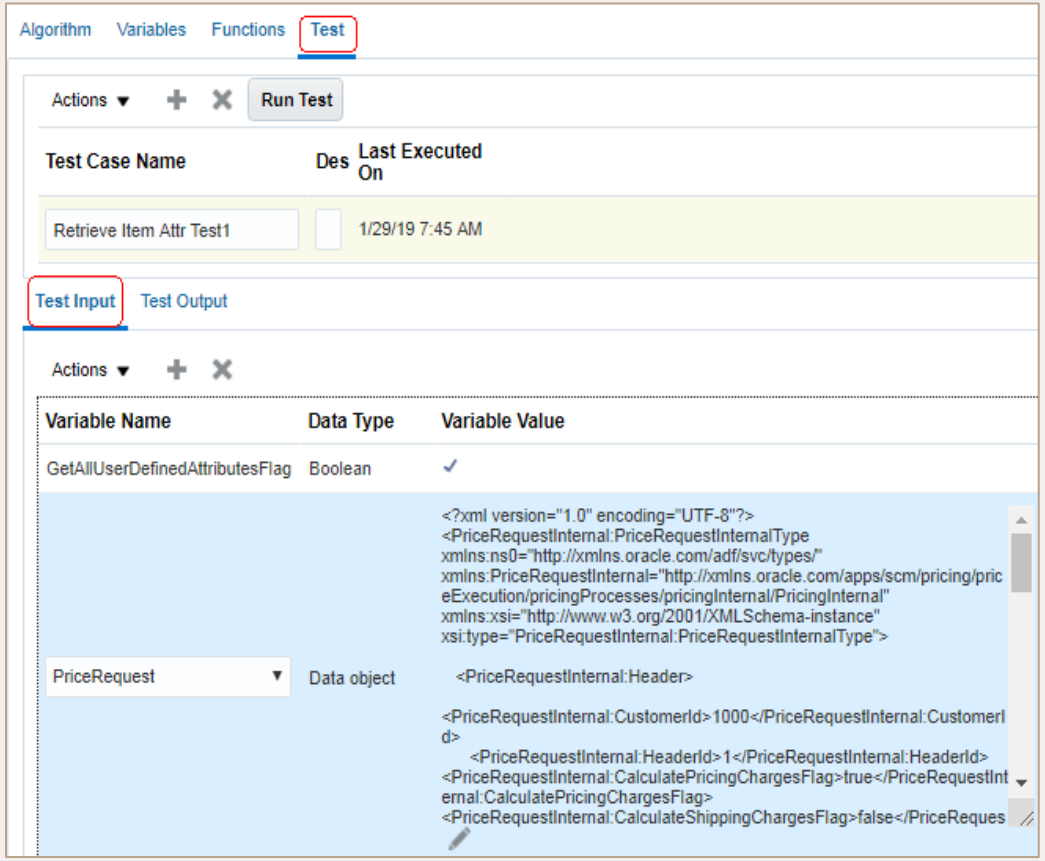
Algorithm Components – Functions View Object

View Object Query Attribute	Description
Application Module	oracle.apps.scm.pricing.priceExecution.pricingProcesses.publicModel.applicationModule.PricingProcessAM
Application Configuration	${(PriceRequest.PricingServiceParameter[0]?.CacheEnabledFlag==null || PriceRequest.PricingServiceParameter[0].CacheEnabledFlag) ? 'PricingProcessAMShared' : 'PricingProcessAMLocal'}
View Object	Eg. CurrencyConvRate1 Or oracle.apps.flex.scm.pricing.priceExecution.pricingProcesses.priceListCharges.view.PriceListChargeDFFVO
View Criteria	Name of the View Criteria.
In Memory Search Specification	Eg. isInDateRange(StartDate, EndDate, pricingDate)
Order By	View Object Attribute to Order By eg. Precedence
Single	Yes/No

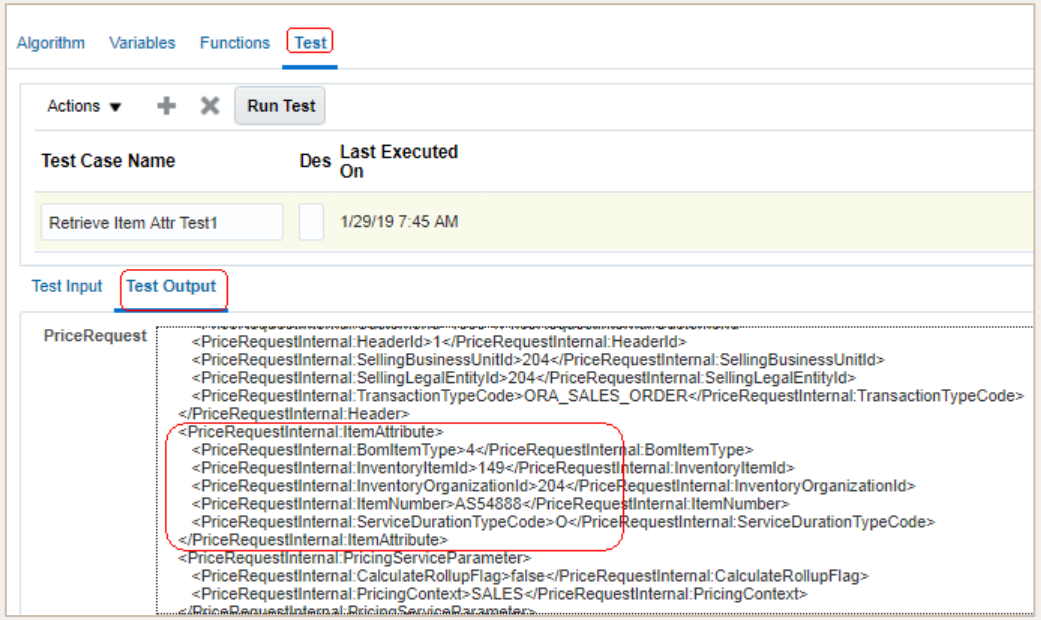
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Algorithm Test Cases

Test Input



Test Output

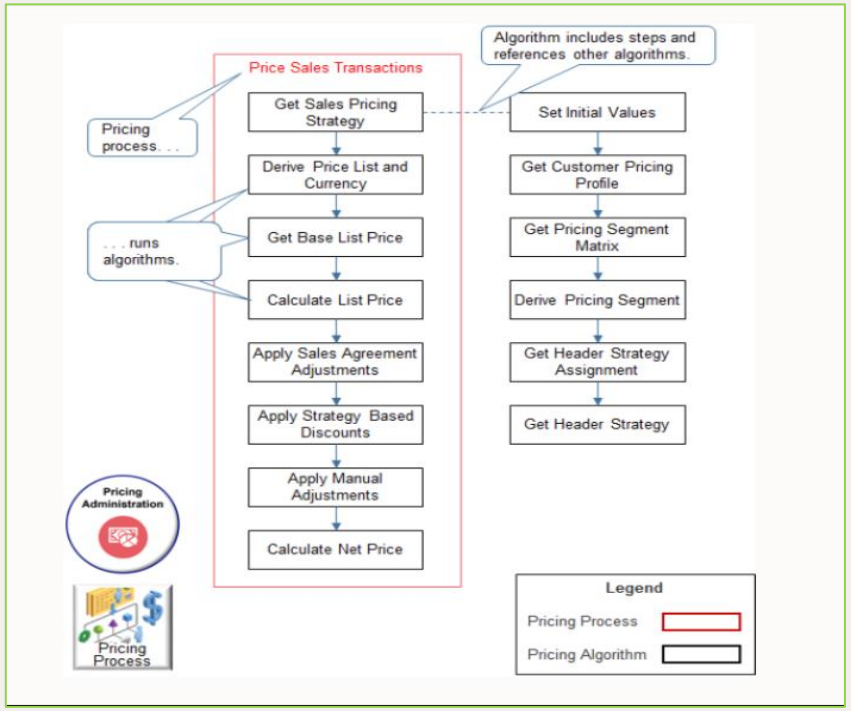


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Pricing Process

A pricing process is an object that runs pricing algorithms to meet the goal of a pricing operation, such as to price a sales transaction.

If needed Algorithms can be customized.



- Pricing comes predefined with a number of pricing processes.
- A pricing process runs pricing algorithms in a sequence.
- The Price Sales Transaction pricing process runs a number of pricing algorithms, such as Get Sales Pricing Strategy.
- Pricing also uses the predefined Calculate Sales Totals and Validate Sales Prices processes.
- Each pricing algorithm contains steps and can reference other pricing algorithms. For example, the Get Sales Pricing Strategy references other pricing algorithms.

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Algorithm Modifications

If you modify the result column of a pricing matrix or use an extensible flexfield with a pricing matrix, then you must modify an algorithm step.

If You Use This Matrix Type	Then Modify This Algorithm Step	In This Pricing Algorithm
Pricing Segment	`Derive Pricing Segment`	`Get Sales Pricing Strategy`
Sales Pricing Strategy Assignment	`Get Header Strategy`	`Get Sales Pricing Strategy`
Price List Charge Adjustment Discount Adjustment Pricing Term Adjustment	`Evaluate Pricing Matrices`	`Apply Matrices`
Line Strategy	`Get Pricing Strategy`	`Get Line Pricing Strategy`
Pricing Charge Guideline	`Evaluate Pricing Matrices`	`Apply Matrix Rules for Guidelines`

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Agenda

- Overview of Pricing
- Pricing Infrastructure Components
- Pricing Matrix
- Service Mappings
- Algorithms
- Use Cases



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Use Case – Use Order Type as a condition in Pricing Strategy Assignment

Assign different Pricing Strategy based on the Order Type. Assume you are a manufacturer of electronic goods and want to provide a few display units to showrooms for demo purposes.

Standard Order	Ship the item at the applicable price
Display Order	Ship the item at zero price

High Level Steps

01	Create the Order Type Code
02	Modify the Service Mapping to add the Order Type
03	Modify the Matrix Class for Sales Pricing Strategy Assignment
04	Setup Pricing Administration to enable the Order Type field and use it in the rules in Manage Pricing Strategy Assignments.

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Use Case – Use Business Unit as a condition in Pricing Strategy Assignment

Assign different Pricing Strategy based on the Business Unit criteria.

	High Level Steps
01	Modify the Matrix Class - Sales Pricing Strategy Assignment to include a condition column for Business Unit
02	Modify the Matrix Class for Sales Pricing Strategy Assignment
03	Setup Pricing Administration to enable the Business Unit field and use it in the rules in Manage Pricing Strategy Assignments.

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Use Case – Adjust the price according to a predefined attribute

Use a predefined attribute like a FOB Code to provide price adjustments on the sales order.

Standard FOB	Ship item at Base Price of say $100
Customer Self Pick	Provide an additional discount of $10, so Base Price of $ 90

High Level Steps

01	Update the Service Mappings to enable the predefined view object FOBPointCode
02	Setup a unique lookup to add the FOB Codes
03	Modify the Matrix Class for Price List Charge Adjustment
04	Setup Pricing Administration to use the FOB Code for setting up required adjustments

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Use Case – Use Item EFF attributes to drive discounting

Use a Item EFF like a Hard disk size to drive differential discounting based on the storage size of the hard disk, which is also stored on the item EFF

All HDD	All HDDs are priced at the same base price of say $ 500
500 GB	If the order is placed for 500 GB HDD, then provide a discount of $ 80
1 TB	If the order is placed for 1 TB HDD, then provide a discount of $ 30

High Level Steps

01	Create the Item EFF and Deploy the Flexfield
02	Create the items and update the EFF with required value
03	Modify the Matrix Class for Pricing Term Adjustment
04	Review and modify the algorithm to include the Item EFF Data set if required
05	Setup Pricing Administration to use the Item EFF Code for setting up required adjustments

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Additional Customization Use Cases

Requirement	Reference
Implement a discount at Order Header level.	Doc ID 2554349.1
Apply a Flat Rate Shipping Charge on an Order. Currently item specific shipping charges are only supported.	Doc ID 2270159.1
Pass Order Management EFF Data To Pricing To Assign Pricing Strategy	Doc ID 2446924.1
Automatically Adjust the Order Line Price Based on a Specific Value From an EFF Field at Order Line Level	Doc ID 2214563.1
Enforce Minimum Order Quantity for an OrderLine, set using PriceList EFF	Doc ID 2662678.1

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New Features

Feature	Release
Import or Update a Large Number of Coverage Items for Price Lists	25B
Redwood: Manage Discount Rules for Component Items	25B
Redwood: Manage Price Lists	25B
Redwood: Use Promotion Codes to Apply Promotions On Order Lines	25B

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New Features

Feature	Release
Evaluate Block Pricing Tiers Across All Lines	25A
Redwood: Apply Promotions According to Items on Sales Orders	25A
Redwood: Create Promotions That Add Free Items	25A
Redwood: Manage Discount Lists	25A
Redwood: Search for Pricing Promotions	25A
Redwood: Use Generative AI to Create Your Pricing Promotion's Description	25A
Use Closed Lines to Help Price Items	25A

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Thank you



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Other Features

Feature	Release
Reset Pricing Algorithms to the Predefined Version	24D
Apply Pricing Promotions to Sales Orders Using Groups of Eligibility and Exclusion Conditions	24C
Create Pricing Promotions for Sales Orders Using Redwood Pages	24C
Target Customers for Pricing Promotions on Sales Orders	24C
Use Categories to Simplify Your Pricing Promotions	24C

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